

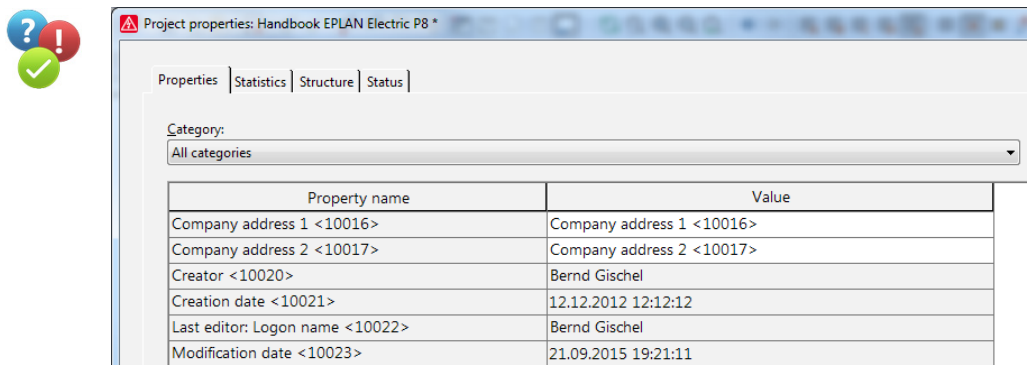


Fig. 2.47 Project properties

The *Project name* <10000> property contains the <Project name> value (the name of the project).

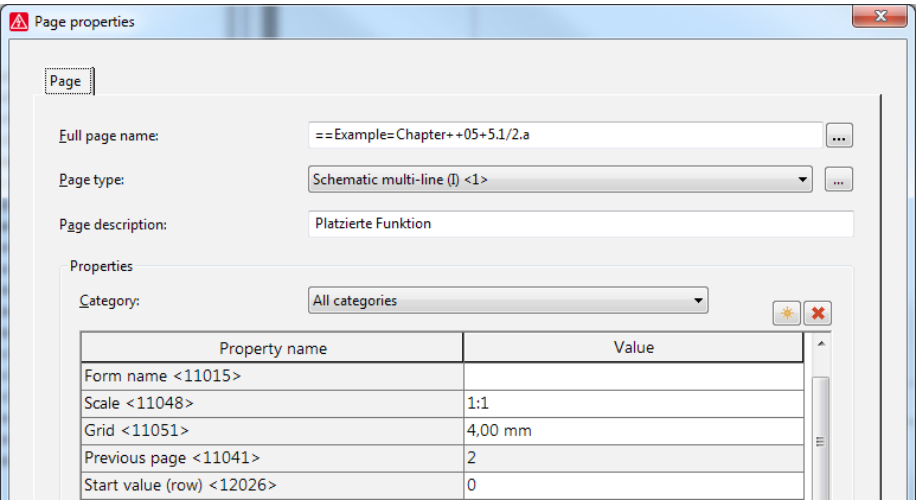


Fig. 2.48 Page properties

The *Page number* <11042> property contains the <n> value (n = the actual page number). EPLAN generally distinguishes between several different types of properties. There are *Project properties* (relevant project information such as the project name; 10000), *Page properties* (properties of a page in the project itself, e.g. the current page number; 11042), *Symbol properties* (property of a symbol such as the displayed device tag; 20010), *Form properties* (properties for constructing a form and reporting data such as terminal and pin designations; 20030), *Plot frame properties* (e.g. the search direction for transferring the device tag, whether the device tag should be automatically transferred from the first device to the left; 12103), and numerous other function properties.

As an addition to the above fixed, not extendable, properties, EPLAN has integrated the user-defined properties that can be edited and configured for your own requirements.

2.5.1 Project properties

Project properties can also be described as global properties. These properties can be used everywhere as special text (project property), in graphical editing for example via the **INSERT/SPECIAL TEXT/PROJECT PROPERTIES** menu item. The **Special text – Project properties** dialog will open.

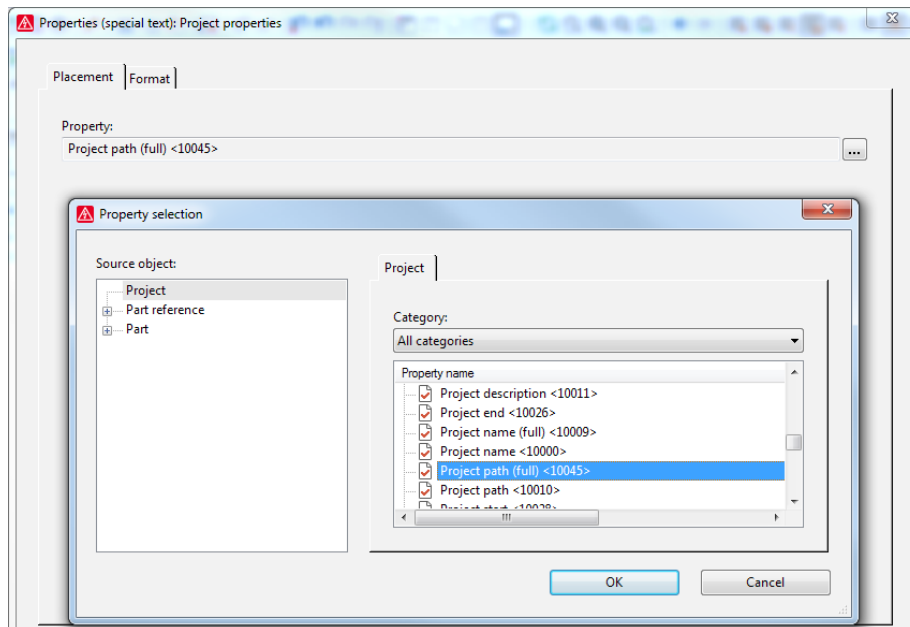


Fig. 2.49 Selecting and inserting a project property

Clicking the  button opens the **Property selection** dialog. Here you can select the desired project property and accept it by clicking OK. The project property is loaded into the *Property* field of the **Special text – Project properties** dialog.

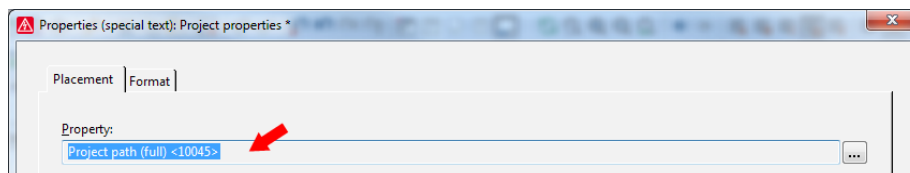


Fig. 2.50 Transferred project property

The special text can then be formatted (font, width, italics, etc.) on the **Format** tab. When you have finished entering data, you confirm by clicking **OK**. The project property now hangs on the cursor and can be placed.

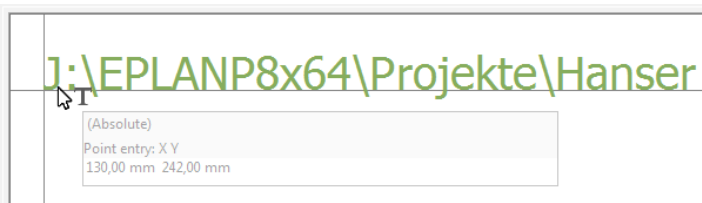


Fig. 2.51 Placing project properties

Project properties can be added to the project via the  graphical button in the project properties. Use the **PROJECT/PROPERTIES** menu to open the project properties. Now click the  button to open the **Property selection** dialog.

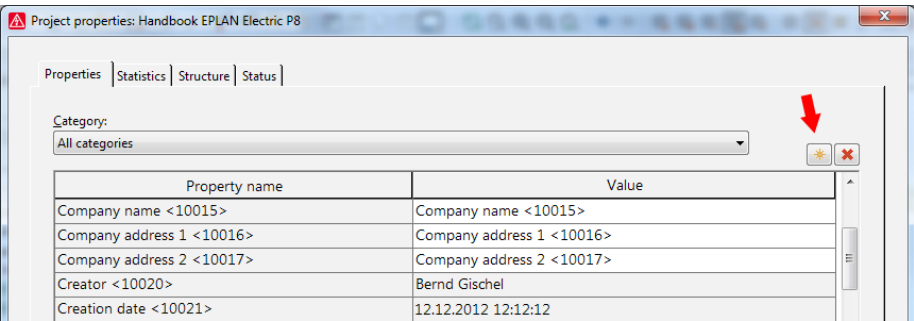


Fig. 2.52 Add project property

In the **Property selection** dialog, the usual Windows functions can be used to select one or more properties and these are then loaded into the project properties of the project by clicking the **OK** button.

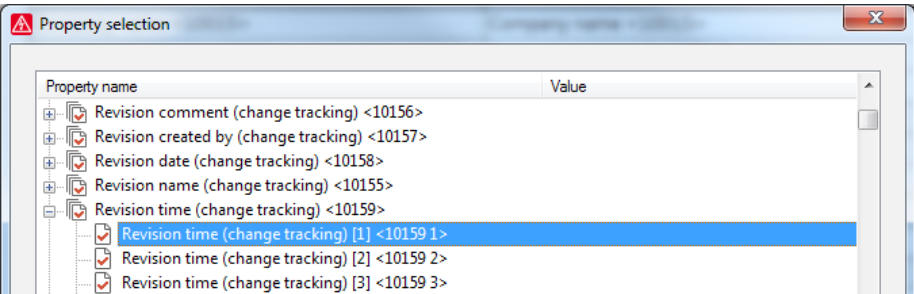


Fig. 2.53 Property selection

You can use the right-click popup menu to sort the display of the project properties. Select the **CONFIGURE** command in the popup menu. The **Property arrangement** dialog opens. The graphical buttons (blue arrows up or down) can be used to arrange the properties as you wish.

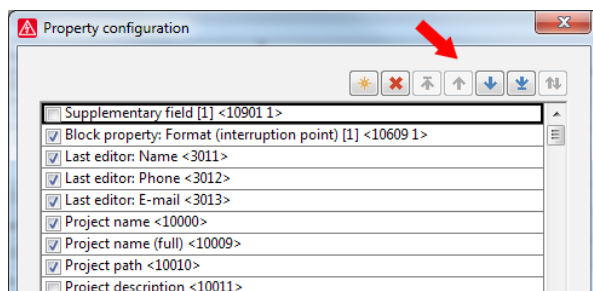


Fig. 2.54 Property configuration

2.5.2 Page properties

EPLAN **page properties** are those properties belonging to a page. Page properties can also be used everywhere. The **INSERT/SPECIAL TEXTS/PAGE PROPERTIES** menu item in graphical editing opens the **Special text — Page properties** dialog. In the *Property* field, you can use the  button to open the **Property selection** dialog.

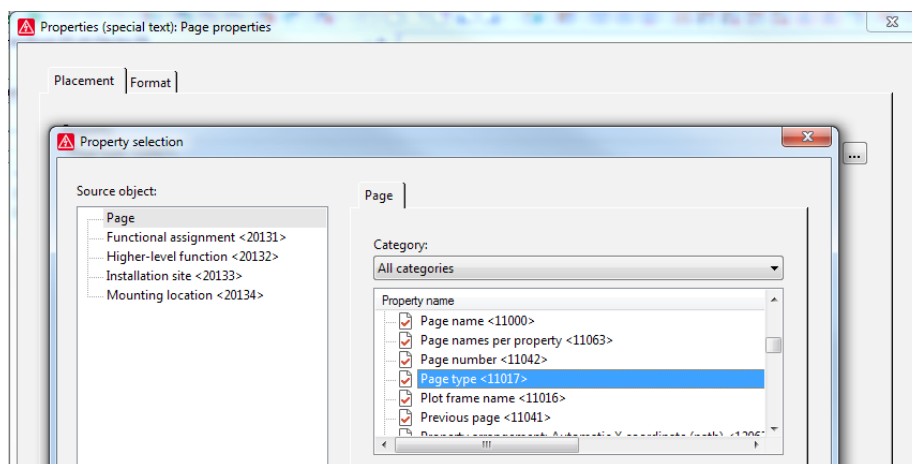


Fig. 2.55 Add page properties

Then you select the desired property. When you click **OK**, the selected property is loaded into the *Property* field of the **Special text — Page properties** dialog.

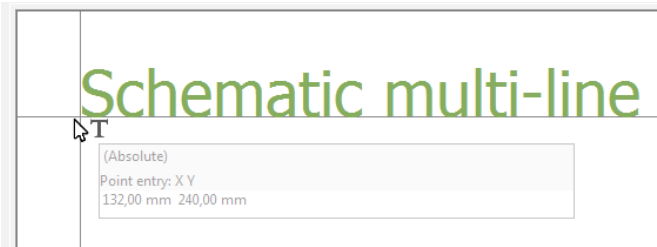


Fig. 2.56
Placing a page
property

After formatting the text (if necessary, on the *Format* tab), you close the **Special text – Page properties** by clicking OK. The special text now hangs on the cursor and can be placed as desired.

2.5.3 User-defined properties

User-defined properties are properties that the user can create and edit freely. User-defined properties are always created in a project-specific manner, but can be easily integrated with or added to other projects by means of EPLAN's integrated import and export functions. User-defined properties are assigned to specific areas (project, page, etc.). Using default selection lists, self-extending selection lists or free entries, they can be created and subsequently selected easily.

User-defined properties are edited using the **CONFIGURE OPTIONS/PROPERTIES** menu.

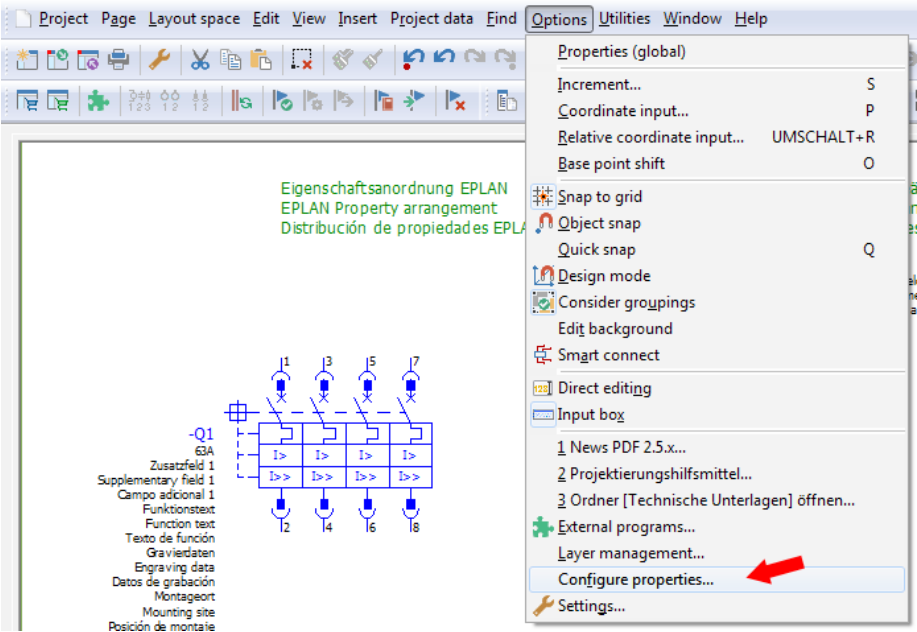


Fig. 2.57
Menu call: configuring
(user-defined)
properties

Afterwards, EPLAN opens the **Configure properties** dialog. If there are no user-defined properties yet, the left area of the dialog will remain empty.

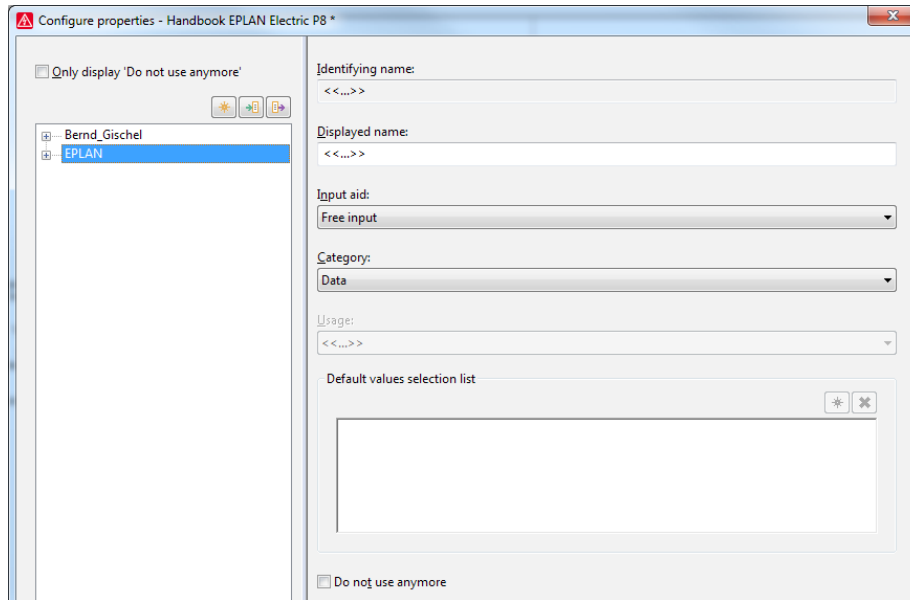


Fig. 2.58 Dialog for configuring properties

Click the **NEW** button (☀ sun symbol) in the left area to create a new user-defined property and configure it for subsequent use. EPLAN opens the **Generate user-defined property** dialog.

Now, the new property can be specified. Depending on the input, EPLAN differentiates several levels. Each new level must be separated by a period.

In the example, the identifying name is thus *Level1*, followed by *Level2*, then *Level3*, etc.

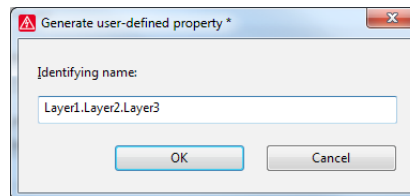


Fig. 2.59
Dialog Generate
user-defined property

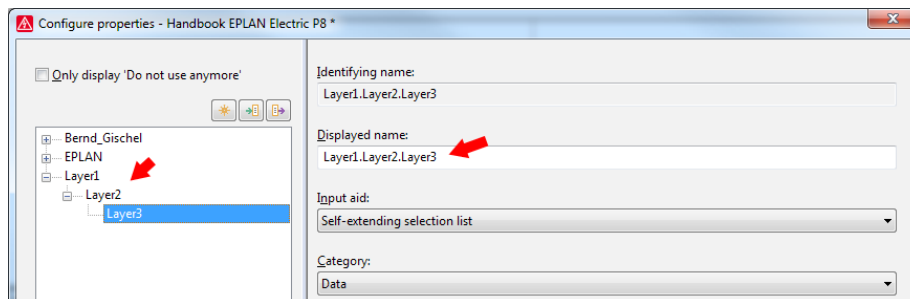


Fig. 2.60 Structure of the identifying name

This is how a structure can be established. Similar user-defined properties are sorted into the structure correctly.

The user-defined property can now be configured further. The following settings are possible:

- **Displayed name:** This can be edited freely and allows for multiple languages
- **Input aid:** Selection of one of the three options (free entry, self-extending selection list, selection list)
- **Category:** Selection of a category (such as devices, data, etc.) to be assigned to the user-defined property
- **Usage:** Selection of objects (project, page, etc.) to which the user-defined property is to be apply

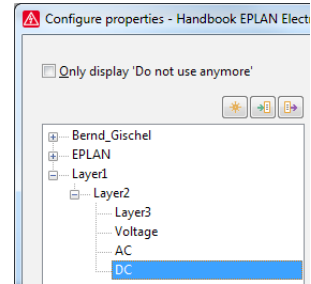


Fig. 2.61
Structure



NOTE: Once the user-defined property has been saved, the Usage selection can no longer be changed.

For example, a selection list (5V DC, 12V DC, etc.) with values can be specified for a user-defined property (control voltage DC) for the Data category and Project (Usage) object to limit the freedom in creating the project.

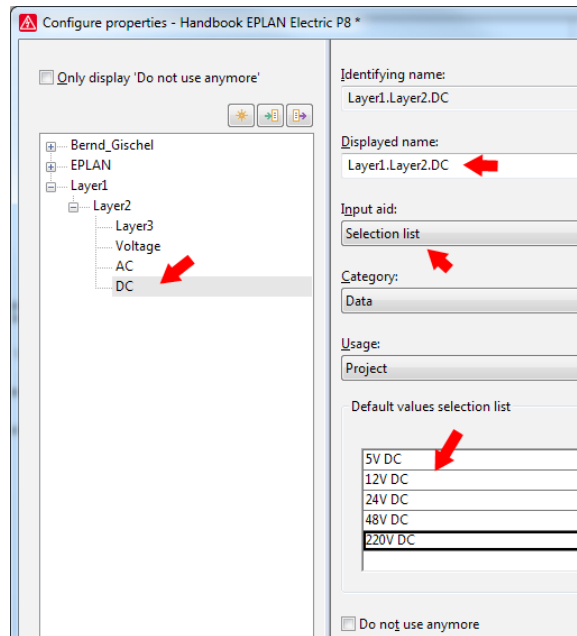


Fig. 2.62 Default values from a selection list

The user-defined property is added via the **NEW** button, as is common EPLAN practice.

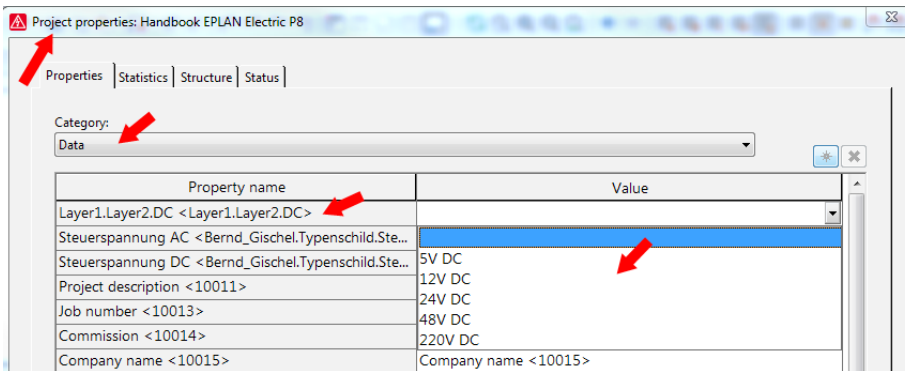


Fig. 2.63 Selection default value

2.5.3.1 Remove user-defined properties

To remove a user-defined property from a project: in the **Configure properties** dialog (CONFIGURE OPTIONS/PROPERTIES menu), select the *Do not use anymore* checkbox for this property.

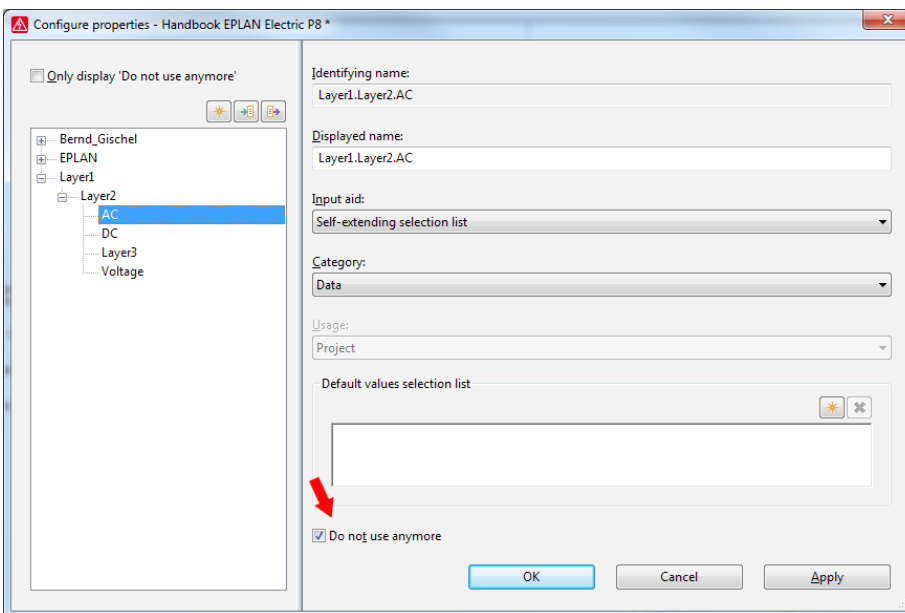


Fig. 2.64 Checkbox Do not use anymore

Entire nodes can be selected as well before selecting the *Do not use anymore* checkbox. These selected user-defined properties can be removed from the project definitively with one compression run.

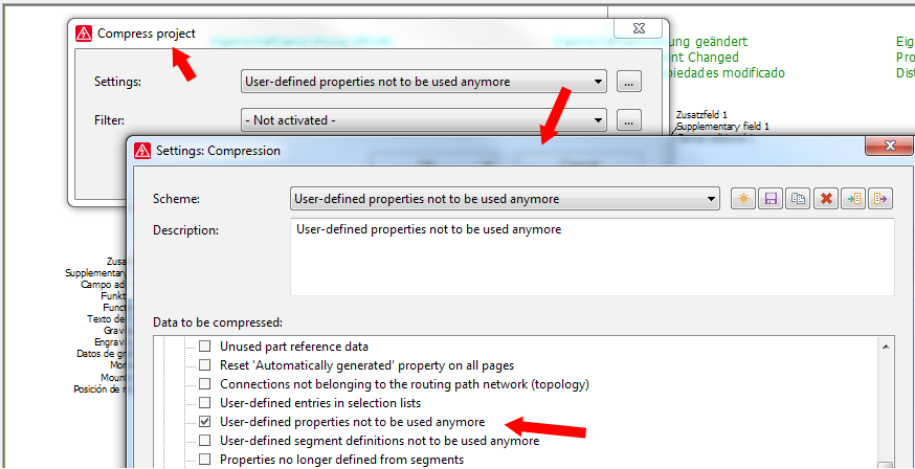


Fig. 2.65 Compression run

2.5.4 Symbol properties (components)

Symbol properties are properties that are assigned to special symbols (components). Some symbol properties, such as connection point designations or device tags, can be accessed directly.

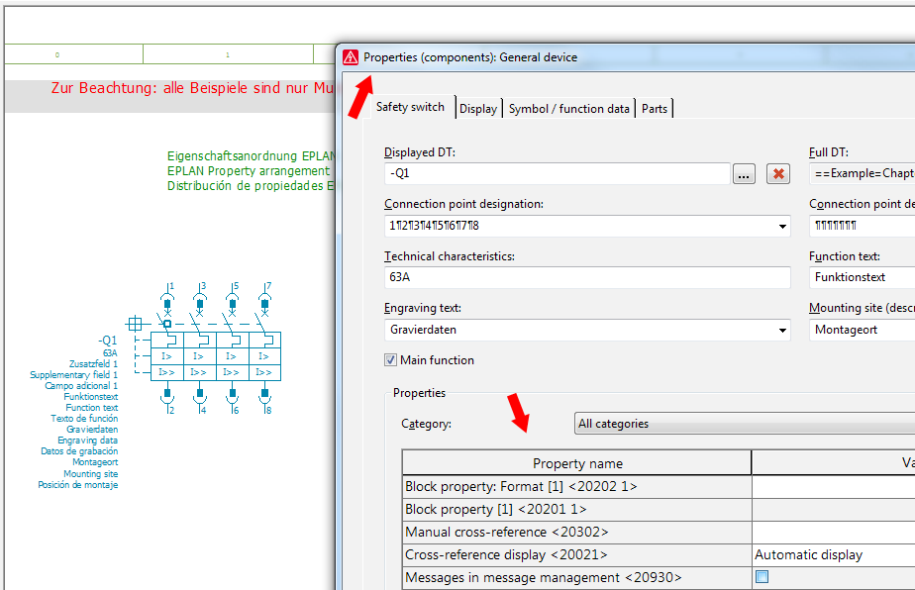


Fig. 2.66 Properties of a symbol (component)

New properties are added in the same way as other properties (project properties or page properties). You click the  graphical button on the *Property (device type)* tab. EPLAN opens the **Property selection** dialog. Here you can select the desired property or properties to be used in the symbol properties and confirm by pressing **OK**. The property has now been added.

It is also possible to adjust the sequence in which properties are displayed in a symbol. You call up the popup menu using the right mouse button. Select the **CONFIGURE** entry in the menu that opens up. EPLAN opens the **Property arrangement** dialog. Here you can now use the familiar graphical buttons to move or sort the properties.

2.5.5 Form properties

In EPLAN, **Form properties** are the properties of a form. Forms (reports) are distinguished by report type. Some properties are shared by all types of forms, but there are also properties that apply only to specific report types. Form properties can be edited only in the **Form editor**.

2.5.6 Plot frame properties

Plot frame properties are properties assigned to the structure and reports of normal page types, such as the *schematic multi-line*. An example of this is the *Path areas* property and its size. Plot frame properties can be edited only in the **Plot frame editor**.

■ 2.6 Buttons and popup menus

EPLAN has many dialogs. This is not unusual in such an extensive program. At this point, I would like to draw special attention to a number of dialog elements that occur repeatedly in many dialogs. These dialogs are similar in many different places in EPLAN. It is therefore important to know what they mean and how they can be used.

Graphical buttons play an important role because they provide easy access to most of the functions and procedures in EPLAN.

2.6.1 Device dialog buttons

Apart from a few small details, all EPLAN device dialogs are very similar. This means that they always have the same structure, regardless of whether it is a device dialog for a motor overload switch or a transformer or any other component. EPLAN uses different buttons in different tabs to simplify operation. Here is a list of the most commonly used buttons and what they mean.

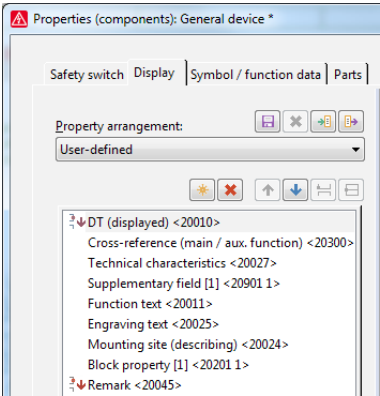


Fig. 2.67
Various buttons

-  Add new property to property arrangement
-  Remove property from property arrangement
-  Move property up one place
-  Move property down one place
-  Undock property (release)
-  Dock property (link to upper area)
-  Save
-  Delete
-  Import
-  Export

2.6.2 Buttons in dialogs (configuring)

You can use the **CONFIGURE** menu item to better arrange or re-sort properties in dialogs to suit your personal needs.

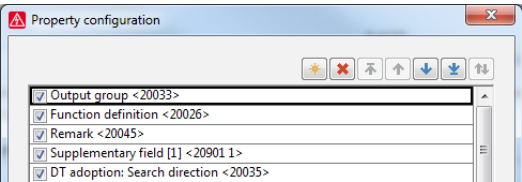


Fig. 2.68
Buttons in the property
configuration dialog

These can be page properties and also settings in other dialogs such as the **UTILITIES/GENERATE REPORTS/SETTINGS/OUTPUT TO PAGES** dialog. The number of buttons can vary here. In principle, the meaning/function of the buttons remains the same.

-  Move the selected entries to the start of the list
-  Move the selected entries to the end of the list
-  Swap two selected entries (swap positions)

In contrast to the graphical buttons in the device dialog, multiple entries can be selected here. The functions provided by the other graphical buttons have already been described.

2.6.3 Buttons in dialogs such as filter schemes

In addition to the already familiar buttons, dialogs such as **Filter**, **Sorting** and other settings dialogs have additional graphical buttons.

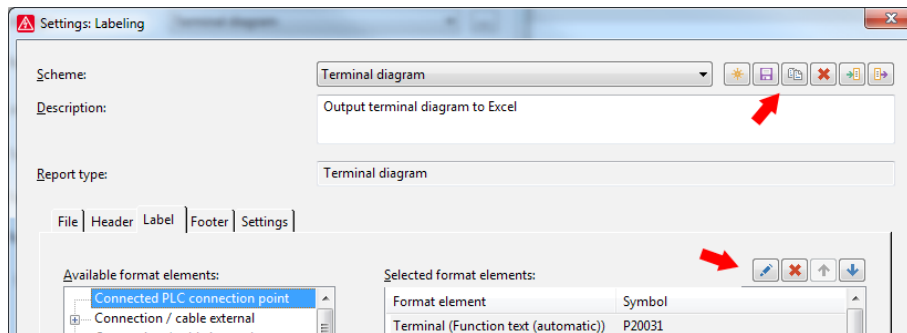


Fig. 2.69 Buttons in various settings dialogs

Additional buttons in the **Filter** or **Sorting** dialogs:

-  Edit entry
-  Copy entry

2.6.4 Restore default function

Depending on the dialog, you may find the *Restore default* function on the right mouse button. To call this function, click the corresponding field (such as Path selection) and then right-click. It is very useful in dialogs that open files or directories. Here you can quickly set the default directories, without having to click all the way through to the default directory.

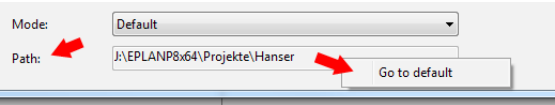


Fig. 2.70 Restore default function

2.6.5 Property arrangements (components)

EPLAN always offers a *Default setting* for property arrangements. The default setting is defined in the symbol structure and cannot be changed here. To change these default settings, you have to edit the symbols. I recommend that you do not make any changes to the symbol libraries that came with EPLAN since changes may make them incompatible with newer versions of EPLAN symbol libraries, making synchronization impossible. You can also change these property arrangements. Once they have been changed, they initially appear in the property arrangement with the designation *User-defined*.

To undo this user-defined property arrangement, simply select the *Default setting* from the *Property arrangement* selection list. This returns all manually changed settings on a symbol back to the default values defined for the symbol.

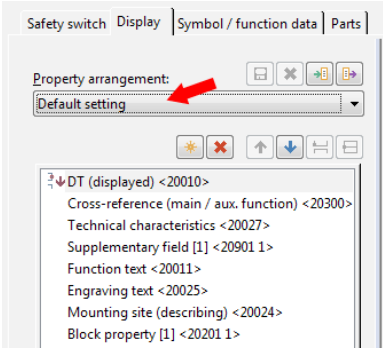


Fig. 2.71
Default setting

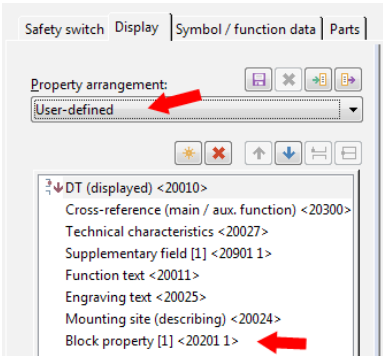


Fig. 2.72
User-defined
arrangement

If a property has been changed, then EPLAN no longer regards it as a default value and the display in the *Property arrangement* selection field changes to *User-defined*. This user-defined arrangement can also be saved via the **SAVE** button and then called up or set at similar symbols (functions). You can assign a descriptive name to the property arrangement. After you click **OK**, this property arrangement is saved, entered into the selection field and is then available to all symbol variants.

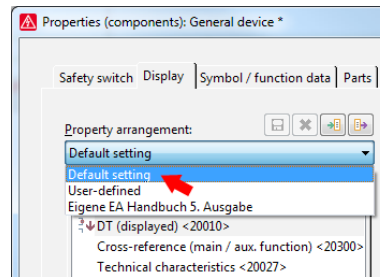


Fig. 2.73
Undo

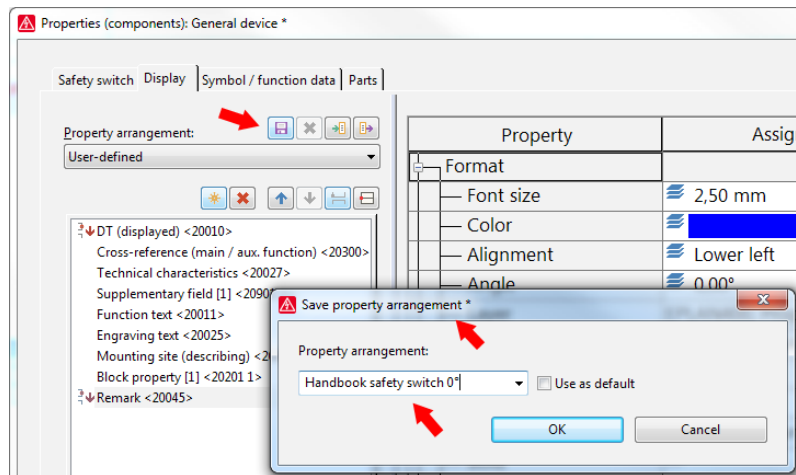


Fig. 2.74 Saving your own property arrangement

Property arrangements can only be transferred to the same “types” of devices (same function template), i.e. the property arrangement of a safety switch cannot be transferred to the property arrangement of a terminal.

The *Use as default* option allows EPLAN to immediately (the next time you insert the same symbol) and automatically use your own property arrangement.

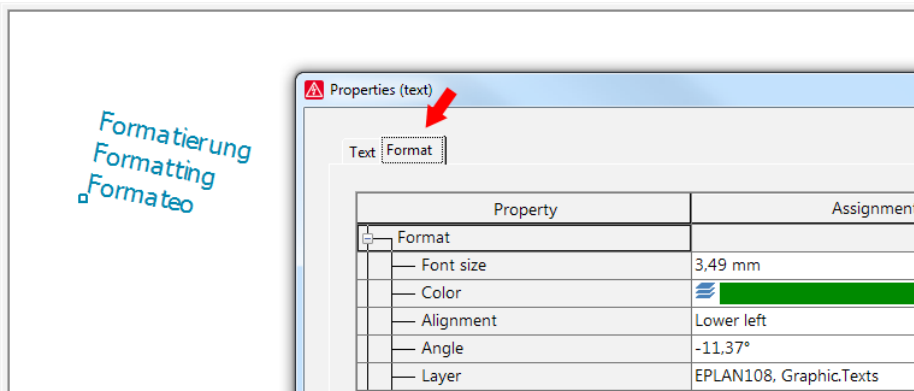
The property arrangements can also be imported or exported in order to continue using them in other projects. Use the **PROJECT/ORGANIZE/PROPERTY ARRANGEMENTS** menu for import/export.

2.6.6 Format properties

EPLAN makes it easy to use different formats, e.g. for texts. Though the dialogs may be somewhat different, the basic principle always remains very similar.

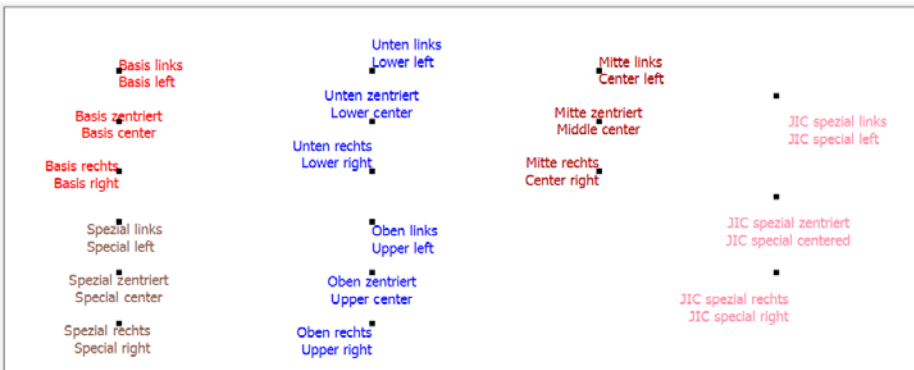
A normal text dialog is used as an example here. To insert a text, you use the **INSERT/GRAPHIC/TEXT** menu to open the **Properties** text dialog.

Fig. 2.75
Formatting options for
texts (example)



You can now enter the desired text into the **Enter text** field on the *Text* tab in the opened dialog. To format the text according to your personal wishes, you need to switch to the **Format** tab. You can freely format the text you just entered. You can edit all selection fields or select default values.

Fig. 2.76
Examples of different
formats



This example shows how different the arrangement and appearance of a text can be if it is formatted using the default setting *Alignment* (the small black dots are the anchor points of the text).

Additional representation in dialogs, including the option of changing formatting (font size, angle, etc.):



This symbol at the start of a line (in the *Value* column) means that the value entered is the original value from the corresponding level and has not been modified. If the value is changed, this symbol will disappear.

2.6.7 Buttons (small black triangles)

In a number of dialogs, the buttons have a small black triangle. Additional menus are “hidden” behind these buttons.

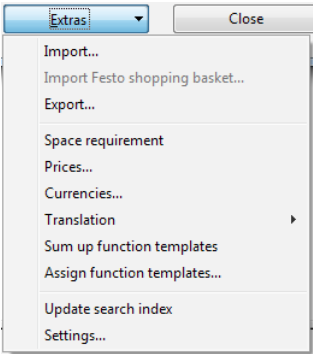


Fig. 2.77 Parts management

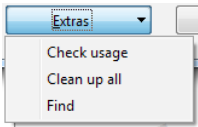


Fig. 2.78 Structure identifier management

2.6.8 Dialogs for schemes

Dialogs for schemes allow you to export or import created schemes.

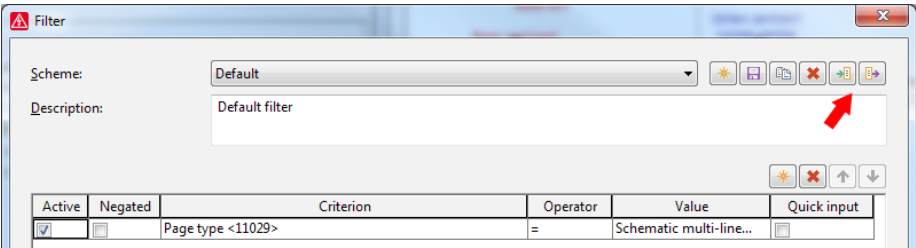


Fig. 2.79 Import and export options for schemes

TIP: User-created schemes should be regularly exported, preferably as soon as they have been created. These schemes are then always available for new versions, other projects or other workstations (depending on the installation).

In many areas, EPLAN offers a special identifier for the scheme type. This way, during an import, only those scheme types that can actually be imported into the current scheme are displayed.



<i>CBnu. *.xml</i>	Example: Scheme prefix for cable numbering
<i>LB. *.xml</i>	Example: Scheme prefix for labeling
<i>WS. *.xml</i>	Example: Scheme prefix for workspaces

2.7 Master data

EPLAN distinguishes between two types of master data: **system master data** and **project master data**.

System master data is stored in the directories with the associated user directory that were set during installation. In addition to the user-specific system master data, the original EPLAN system master data is also installed in the *EPLAN original master data directory* (depending on the directory selected during the installation).

It should be noted that on installation, only the original EPLAN system master data is overwritten or updated in the *\EPLAN* directories. In case of a new installation or installation of an update, the user-specific system master data is **not** overwritten or updated. Generally, though, it is possible to synchronize the user's own system master data with the original EPLAN system master data. To do so, in the **UTILITIES/MASTER DATA** menu, you call up the **SYNCHRONIZE SYSTEM MASTER DATA** menu item. In the following dialog, you can then update your own system master data accordingly.

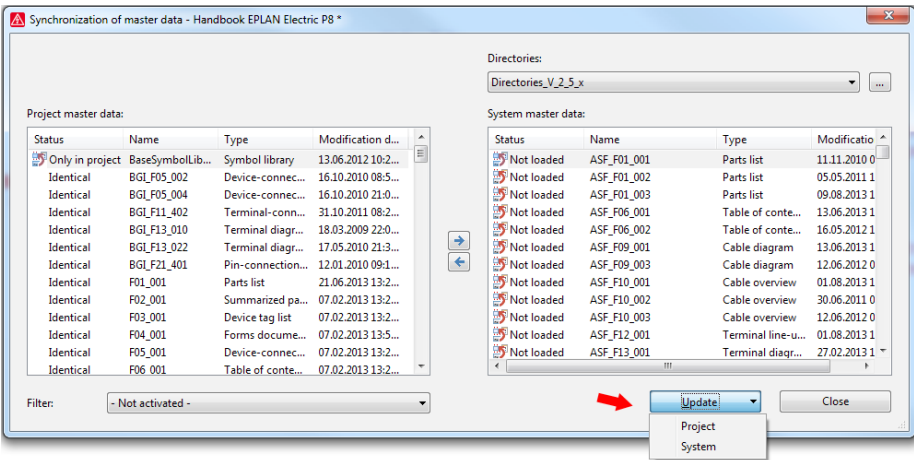


Fig. 2.80 Master data synchronization

If, for example, you have modified the supplied table of contents form *F06_001.f06*, EPLAN would not replace this form in the user-specific master data directories `\\Forms\Company code` with an updated, possibly improved, version. It will only overwrite this form in the `\\Forms\EPLAN` directory.

System master data contains data such as symbol libraries with the associated symbols, function definitions, forms and plot frames.

Project master data is the other type of master data. Project master data, after initial use, is moved from the system master data and stored in the project; after it is stored it is independent of the system master data.

■ 2.8 Operation

EPLAN can be conveniently operated using the mouse. However, to increase your own working speed, EPLAN is very flexible in allowing functions accessed with the mouse to also be accessed via definable keyboard shortcuts. Normally functions can be assigned to a keyboard shortcut if they are also accessible in the main menus, such as Page, Project etc.

To translate text in a Text dialog, you can select the appropriate command with the mouse via **UTILITIES/TRANSLATION**. This function can be assigned a keyboard shortcut because it is a normal menu command. One possibility is the key combination **CTRL + T**.



2.8.1 Using the keyboard

EPLAN allows the user a great deal of freedom in assigning keyboard shortcuts to functions and commands.



NOTE: This is one limitation here: Certain shortcuts are often only permitted in combination with the CTRL and/or ALT keys.

Of course, important standard Windows shortcuts such as **CTRL+ C (COPY)** or **CTRL + V** (paste) should **not** be reassigned if at all possible. It is possible, but not recommended.

Selected shortcuts

Here are a number of recommended keyboard shortcuts that have proven useful in practice.

Copy page from /to	CTRL + SHIFT + P
Edit /Copy format	ALT + F
Edit /Assign format	ALT + A
Utilities /Reports /Generate	CTRL + ALT + G
Find /Synchronize selection	CTRL + SHIFT + Y
Project data /Device navigator	STRG + SHIFT + D
Project data /Update connections	F11
Utilities /Parts /Management	ALT + V
Utilities /Translation /Dictionary	ALT + T

To assign your own shortcut to functions, in the **OPTIONS/SETTINGS** menu, select **USER/MANAGEMENT/SHORTCUT KEYS**.

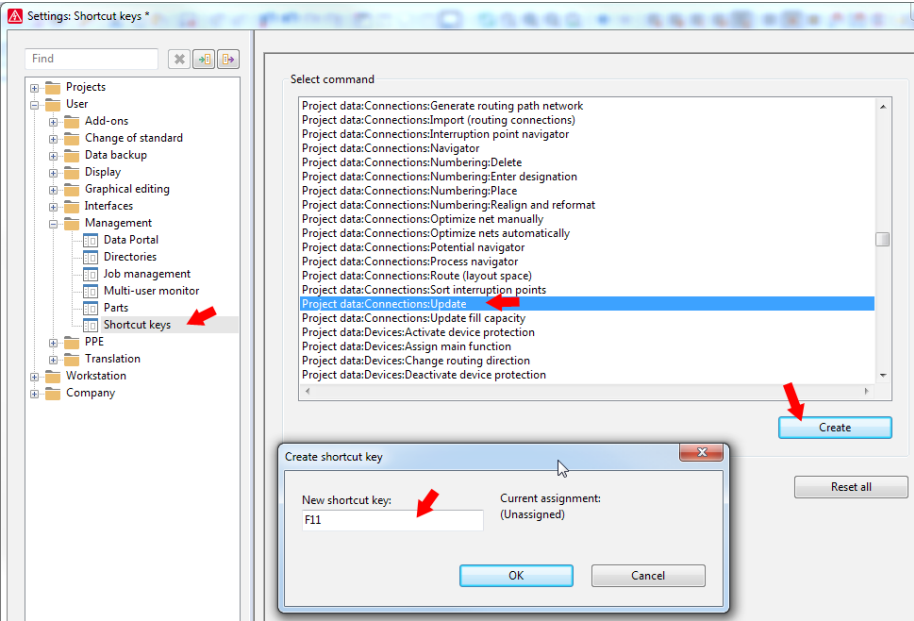


Fig. 2.81 Assigning a shortcut key to a function

In this dialog, you select the desired action and assign it a shortcut key using the **CREATE** button. The shortcut key is applied and fully “operational” as soon as you click **OK**.

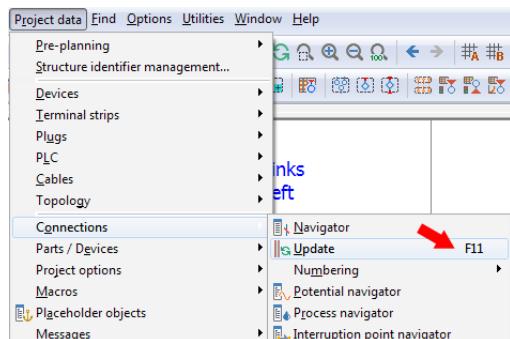


Fig. 2.82

Assigned shortcuts in the Connections menu

2.8.2 Using the mouse

EPLAN can be fully operated using the mouse. EPLAN can be completely operated using the mouse. Admittedly, many menu items are easier to reach with the mouse than with cryptic key combinations that one usually cannot remember.

■ 2.9 User interface – more useful information

EPLAN provides a great deal of support to the user when editing projects, but also for general operation of the program.

A very nice and practical feature is the ability to freely configure the EPLAN user interface. The so-called workspaces are defined depending on the logged-in EPLAN user. This makes it possible to always call up your own user interface configuration when you start up EPLAN on the same computer.

2.9.1 Using workspaces

Put simply, a **workspace** is the dialog layout and the toolbars you want to have when performing certain operations in EPLAN. EPLAN allows you to create separate workspaces for particular areas that contain exactly these desired toolbars, views or dialogs.

The Workspace function is accessed from the graphical editing menu via **VIEW/WORKSPACE**. When you select this function, EPLAN opens the **Workspace** dialog and you can use the selection field to select one.

2.9.1.1 Creating and selecting a workspace

A workspace is customized by placing desired dialogs or toolbars to suit your personal preferences. To permanently save a new workspace, you open the **Workspace** dialog via the **VIEW/WORKSPACE** menu.

A new workspace is created via the  button. EPLAN then opens the **New scheme** dialog. You should enter a name not yet assigned in the *Name* field and a sensible description for the workspace in the *Description* field.

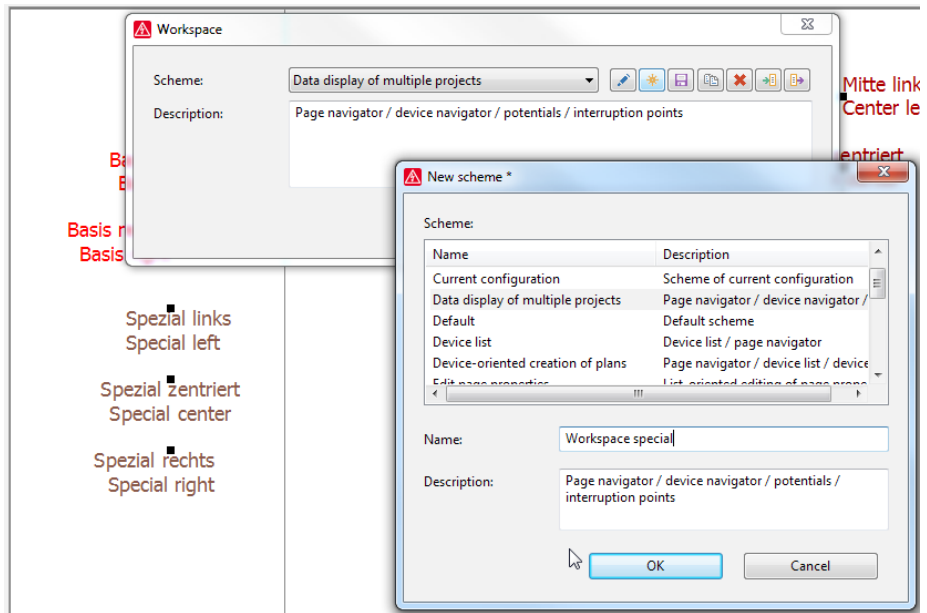


Fig. 2.83 Creating a workspace

When you click **OK**, the workspace is saved and can then be set as the current workspace.

2.9.2 Dialog display

EPLAN provides a lot of data and additional information. EPLAN has a number of different dialog representations in the navigators to ensure information is clearly displayed.

2.9.2.1 Tree view

The **tree view** is one type of representation. Similar to Windows Explorer, all information is displayed in a tree with small symbols.

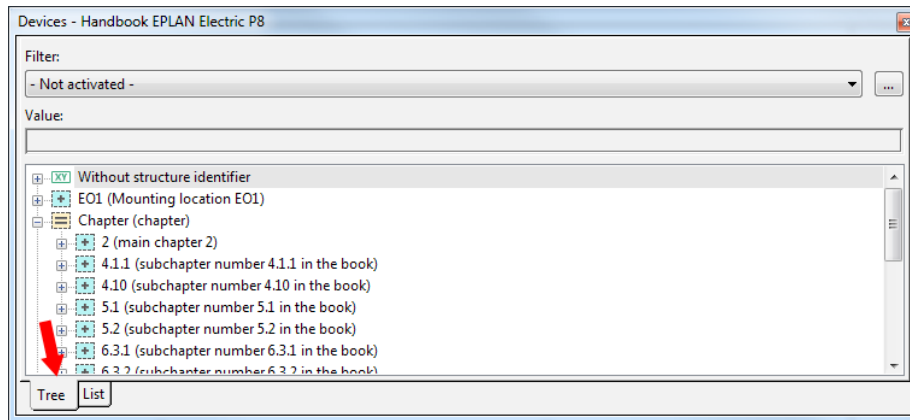


Fig. 2.84
Tree view

This representation type provides a clear overview, making it relatively easy to find objects. In this representation, EPLAN always shows all devices (depending on the function definition). No changes to the form and content of the tree view are possible. EPLAN provides no options for this here.

The display sequence of the devices can be configured under **OPTIONS/SETTINGS/PROJECTS [PROJECT NAME]/DISPLAY/PROJECT STRUCTURE (NAVIGATORS)** and/or **PROJECT STRUCTURE (PAGES)**. This is where you can define how devices should be shown in the tree view, for example, by identifier or by the page prefix.

2.9.2.2 List view

The **list view** is another representation type. This representation type offers many more customization options to suit your working habits. As with the tree view, EPLAN shows all devices here as well (again depending on the navigator that was selected). The cable navigator shows only cables, the terminal navigator shows only devices with the terminal function type, etc.

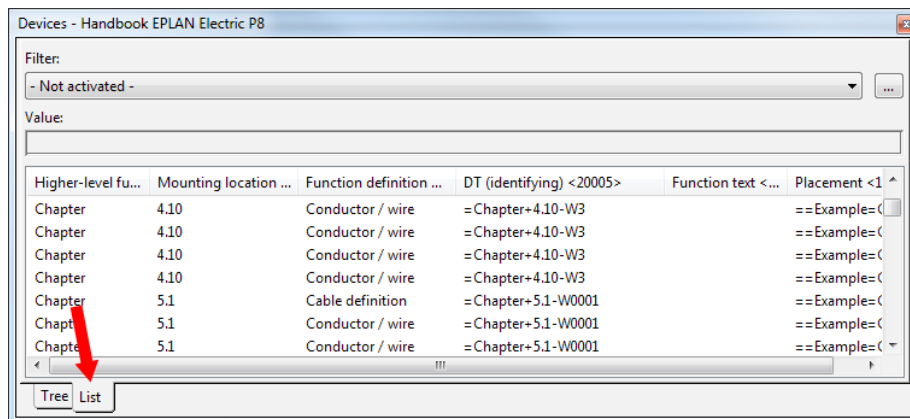


Fig. 2.85
List view

Unlike the tree view, it is possible to add further information to the list view. Right-click to call up the popup menu to select the *Configure representation* function. EPLAN opens the **Configure representation** dialog. You can now select or deselect columns and change their order. Click the OK button to save the settings. You now have a personalized list view.

2.9.2.3 Combined tree and list view (part)

In addition to the previous representations, the *Part master data* navigator also allows a combined display of list and tree view, the **combination** view.

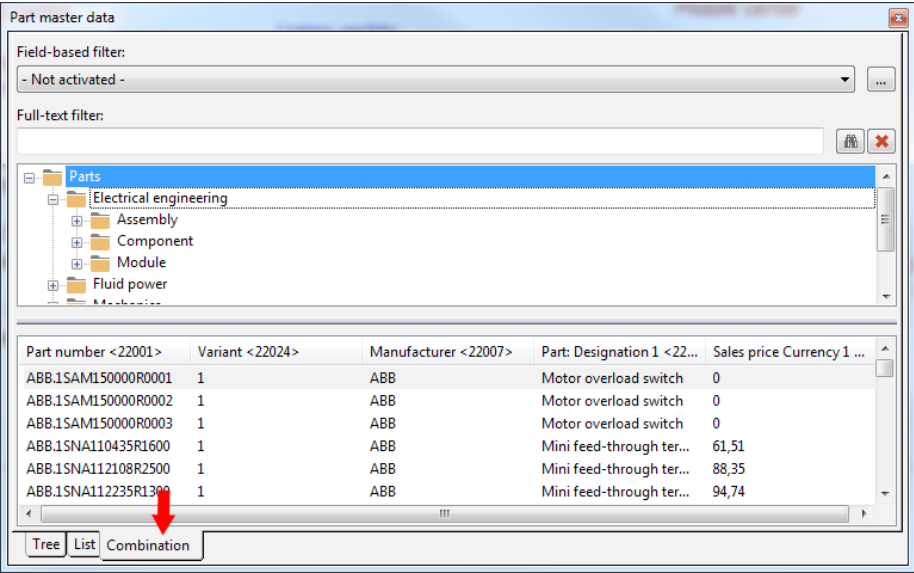


Fig. 2.86 Combined display

The tree view is displayed in the upper area. Here it is possible to preselect, for instance by selecting the *Cable* node. In the lower area, the preselected parts are shown in a clear list with additional specific information.

2.9.2.4 Edit in table

Edit in table is another method of conveniently displaying and editing properties. Edit in table is usually accessed from the tree view by right clicking to access the popup menu. Unlike the previous representations (tree or list view), only the devices and their functions that were previously selected in the navigator are displayed.

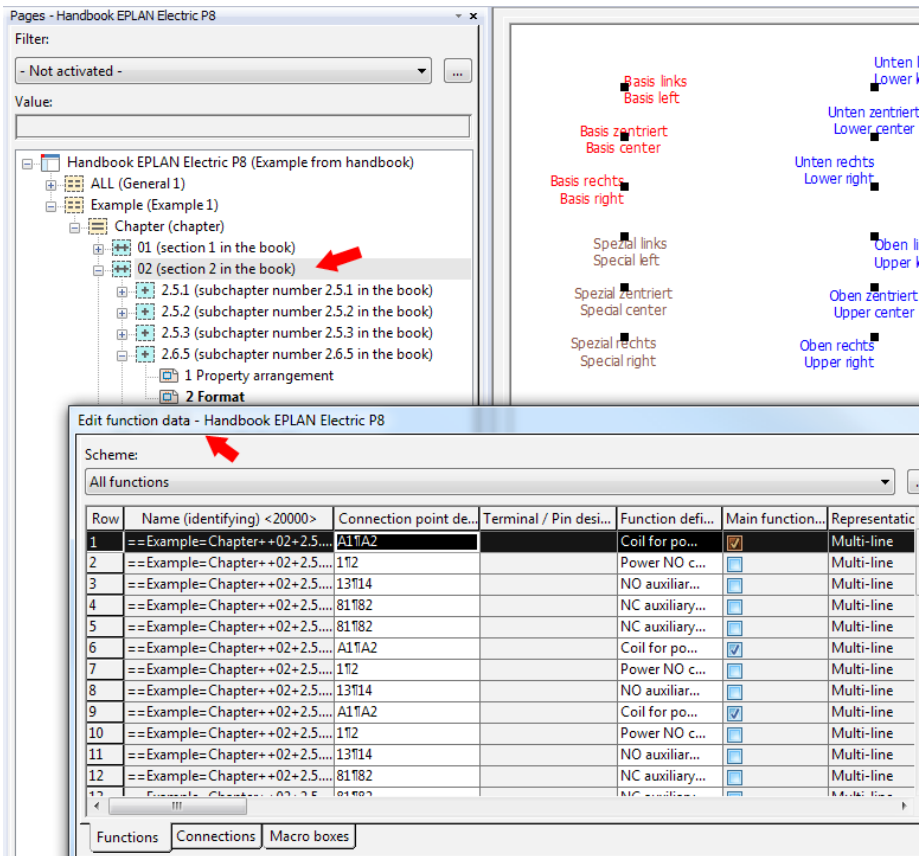


Fig. 2.87 Editing function data in table

The displayed devices can now be edited in a table in the **Edit function data** dialog. All editing functions, such as Copy, Paste and Edit, can be used when editing tables.

3

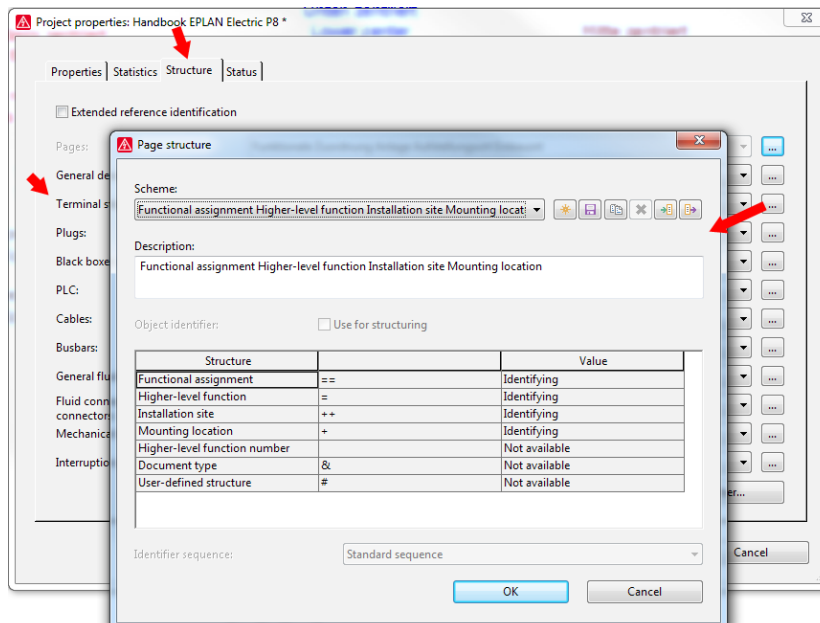
Projects

A project must first be created before you can edit schematics in EPLAN.



NOTE: In EPLAN, the storage location of projects and the names of the individual projects can be freely chosen. The only limitations are the naming conventions of the Windows system being used.

When the page and device structures are defined, a project has all the properties it needs, such as the *device structure* (the structure identifier, e.g. higher-level function or installation site, that is to be used) and, during the time a project is edited, also all the master data, such as forms, parts data, function definitions, symbol libraries, etc., that is necessary for editing projects.



Global structure
of a project

Fig. 3.1
Project properties –
Structure tab

This data is all **completely** stored in the project. Data added later, such as newly used forms, is also stored in the project.

This ensures that this project can later be edited with exactly the same data used when the project was created, or which was generated at the beginning of project editing, or was later stored in the project.

■ 3.1 Project types

EPLAN basically distinguishes between two **project types**: a *schematic project* (this is the usual project in practice) and a *macro project*.

A *macro project* is used for creating and automatically generating window macros, and, using the macro navigator, for managing macros. Logical functions such as cross-references or connection information are not supported in a macro project nor are they displayed.

If necessary, a *schematic project* can easily be modified by changing the <10902 Project type> property from *schematic project* to *macro project*.

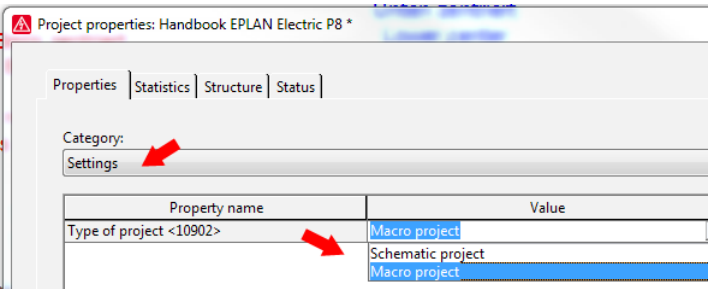


Fig. 3.2
Project properties –
Properties tab

Other project types such as form project or symbol project do not exist. Master data (e.g. forms or symbols) is always temporarily edited directly in the project (always with existing system master data), stored after the form is closed, and then possibly automatically (depending on the settings) synchronized with the current project (depending on the settings and usage) and with the project master data.

And of course the system master data can also be manually synchronized with the project master data. It is also possible to synchronize in the other direction, to synchronize the project master data with the system master data. If EPLAN finds inconsistencies in the master data at this point, then a message is generated and EPLAN cancels the synchronization because incompatible master data cannot be used to overwrite existing master data. This is the case, for example, when connection points are added to existing, placed symbols. The logical data of this symbol no longer “fits” already placed or existing symbols and therefore this data cannot be synchronized automatically anymore.

EPLAN manages different project types as schematic projects.

3.1.1 Project types in EPLAN

- Editable EPLAN projects (*.elk;*.ell)
- Editable EPLAN projects (*.elk;*.ell)
- All EPLAN projects (*.elk;*.elp;*.els;*.ebx;*.elr;*.ell;*.elt)
- EPLAN project (*.elk)
- EPLAN project with change tracking (*.ell)
- Zippped EPLAN project (*.elp)
- Filed-off EPLAN project (*.els)
- Filed-off, zippped EPLAN project (*.ebx)
- Completed EPLAN project (*.elr)
- Temporary EPLAN reference project (*.elt)

Project types define the project by its functional meaning, for example, as a normal project or a revision project. EPLAN distinguishes between the project types described in the following section.

- ***.elk:** a normal editable project (normal EPLAN schematic project)
- ***.ell:** a project with change tracking
- ***.elp:** a zipped project
- ***.els:** a filed-off project
- ***.elx:** a filed-off and zipped project
- ***.elr:** a completed project
- ***.elt:** a temporary EPLAN reference project (comparison)

Central project editing occurs in a normal project. All other types of projects (e. g. a completed project, an archived project, a basic project, or a project template) are derived from the normal EPLAN project (schematic project).

3.1.2 Project templates and basic projects

EPLAN allows the user to quickly and precisely create new projects based on existing *basic projects* (*.zw9) and *project templates* (*.ept).

A project template contains preconfigured values. Later, when using the template for a new project, the project and page structure can be change **one time**.

Opening a project template

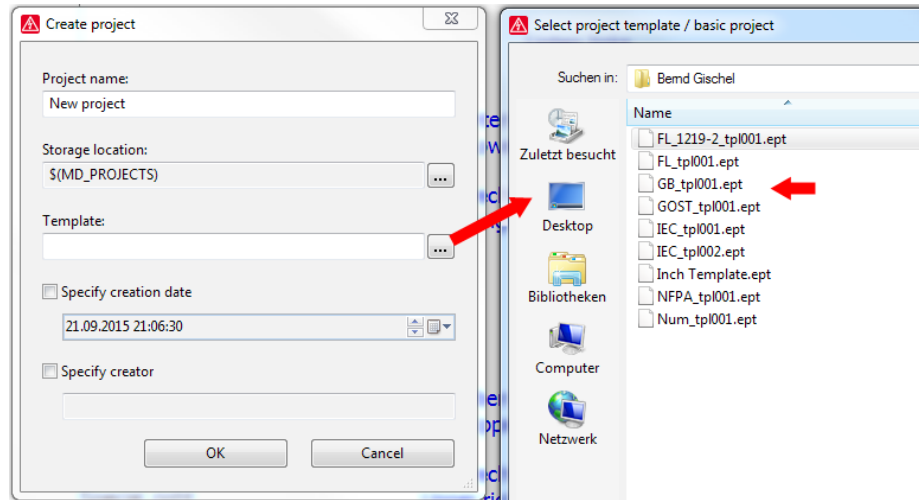


Fig. 3.5
*.ept project template

In contrast to a basic project, a new project created from a project template (via **PROJECT – NEW**) usually has no pages because the page structure still has to be defined. But project templates can also contain pages.

A *project template* has the ***.ept** file extension and cannot be directly opened or modified. You can, however, easily overwrite an existing project template that has incorrect or changed properties with a new project template.



NOTE: Project templates with the *.epb file extension are old versions of project templates, which can no longer be generated. But they can continue to be used.

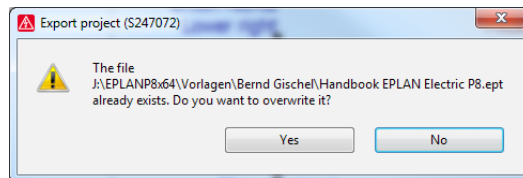


Fig. 3.6
Overwriting a project template

Project templates can be generated from existing projects in the graphical editor via the **PROJECT/ORGANIZE/CREATE PROJECT TEMPLATE** menu. In the page navigator, you first use the mouse to select the project that is later to become a project template and then open the **PROJECT/ORGANIZE/CREATE PROJECT TEMPLATE** menu.

In the **Create project template** dialog that is displayed, you need to define the storage location and the name of the new project template. It is a good idea to create a *project templates* directory below the root directory, possibly with separate folders for each customer.

Basic projects are, for example, projects prefilled with appropriate customer values such as a predefined page structure, sample pages, graphical report templates, various master data, and much more.

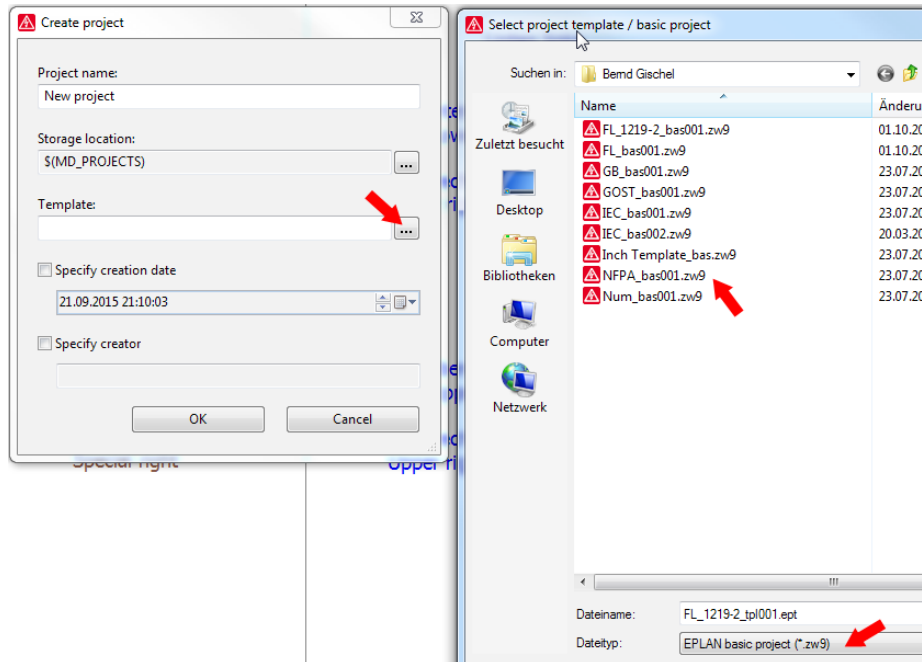


Fig. 3.7 Opening a basic project

In a project created from a basic project, the project and page structures are fixed and can no longer be modified.

A basic project has the *.zw9 file extension and, like project templates, cannot be directly opened. Basic projects, like project templates, can be generated from existing projects in the graphical editor via the **PROJECT/ORGANIZE/CREATE BASIC PROJECT** menu.

EPLAN then executes a number of functions to create the basic project. Afterward, the **Create basic project** dialog opens and this is where you define the directory and the project name of the basic project to be created.

Here too it is a good idea to create a **basic projects** folder below the root directory, possibly with separate customer folders.

Once basic projects or project templates have been created, they cannot be later changed (i.e. directly edited). However, they can be overwritten with new or modified data, by specifying the same file name.

■ 3.2 Creating a new project

To quickly get started with a project, new projects can be generated in various ways in EPLAN. There are two ways of doing this directly from *project editing*.

A new project based on an existing basic project or project template can be created “at the push of a button” via the **PROJECT/NEW** menu.

EPLAN comes with several basic projects and project templates. The project template *IEC_tpl001.ept*, for example, contains the IEC identifier structure, while the supplied project template *Num_tpl001.ept* contains a sequential numbering structure.

Another way to create a new project is via the optional project management. A new project based on an existing basic project or project template can be created here just as fast as via the **PROJECT** menu.

Of course it is also possible to simply copy a project and give it a different name.

3.2.1 New project (from a basic project)

You can either select the menu item **PROJECT/NEW** or use your own keyboard shortcut to create a new project.

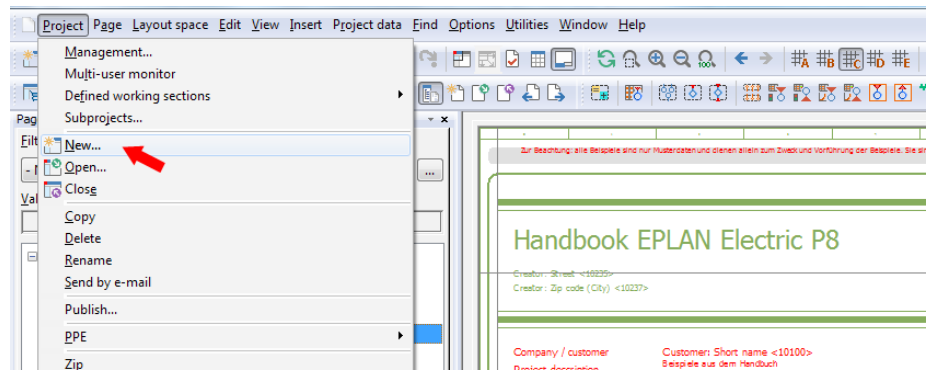


Fig. 3.8
Project/New menu

The **Create project** dialog is displayed. In this dialog, you must/can define the following settings:

- **Project name** input field: The new project name (you need not worry about file extensions; these will be automatically added by EPLAN) is entered here.



NOTE: The project name is not checked to see if it already exists until a template (project template or basic project) has been selected and the Next button has been clicked. The action can then still be cancelled.

EPLAN always suggests the project name New name, possibly also with a consecutive number. This can, of course, be changed to any other name.

- **Storage location** selection field: Specifies the storage location of the new project. Any storage location can be selected using the  button.
- **Template** selection field: You select a template using the  selection button. The usual range of templates and basic projects are available for selection.
- Input field/selection field **Specify creation date**
- Input field **Specify creator**

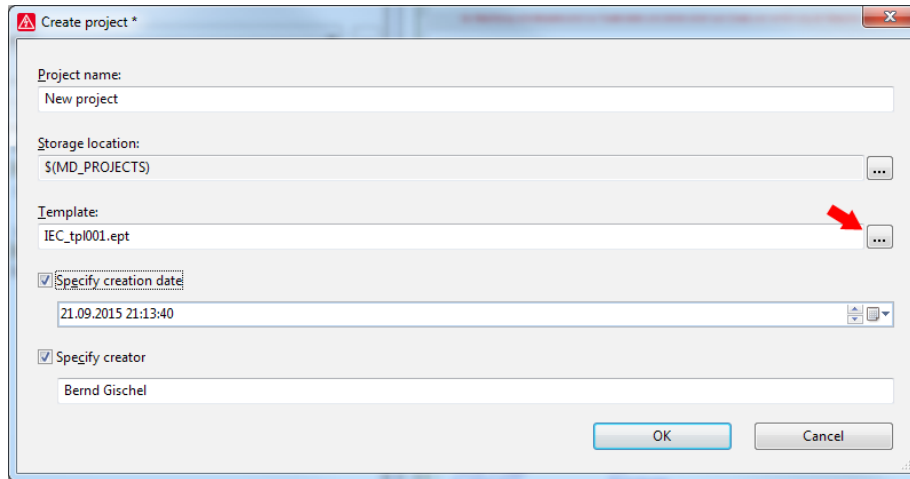


Fig. 3.9 Create project dialog

You can, but do not have to, fill in the information under *Specify creation date* and *Specify creator*.



NOTE: These fields cannot be changed later on. It is therefore very important to make sure you enter the correct data.

After defining the above entries, you need to select a template. This can be a project template or a basic project.

You simply click the More button. In this example, the **Select project template/basic project** dialog opens, and the basic project **Handbook.zw9** will be used to create the new project. You can select the relevant project type in the *File type* field.

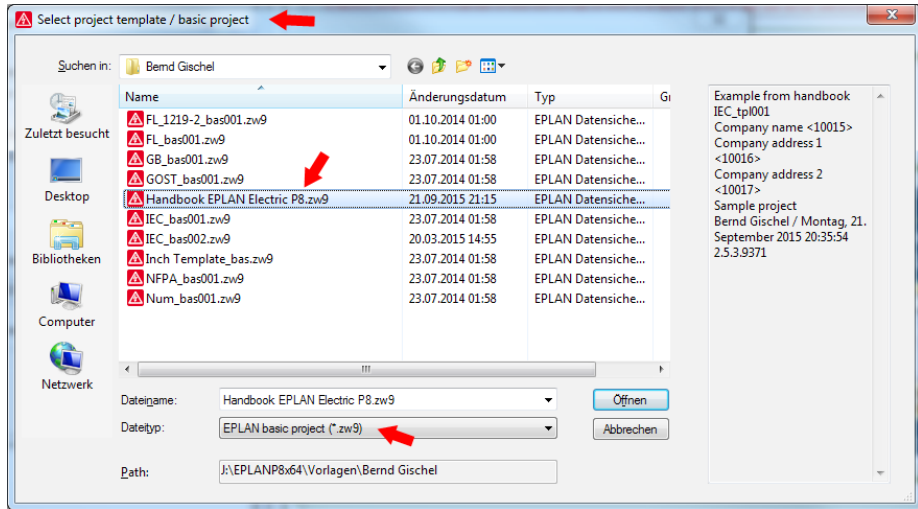


Fig. 3.10 Select project template/basic project dialog

Creating a new project based on a basic project

In the **Select project template/basic project** dialog, you now press the **OPEN** button and select and use an existing basic project as your template.

EPLAN then returns to the **Create project** dialog and imports the selected basic project (or the template project) into the *Template* field.

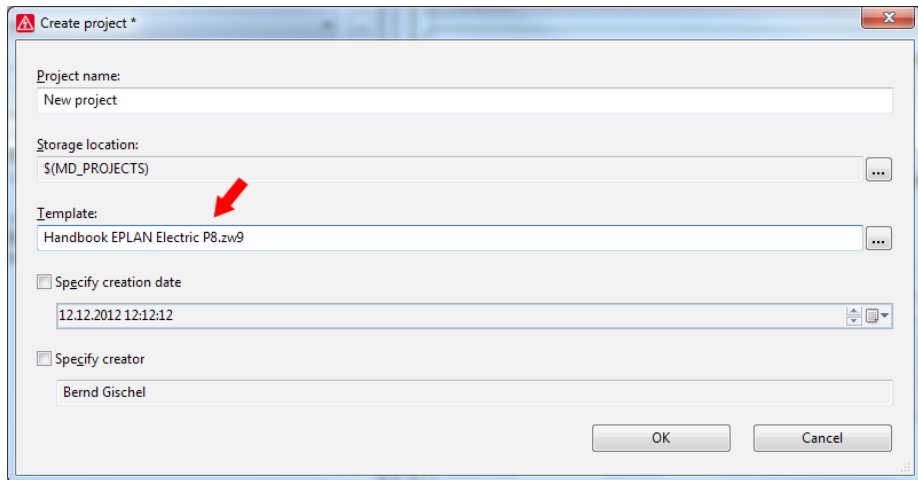


Fig. 3.11 Create project dialog, with the template

Now, you only have to confirm this dialog by clicking **OK** and EPLAN will generate the new project in the specified directory.

The new project is created from the selected basic project. This may take a while depending on the hardware and the storage location (server, local).

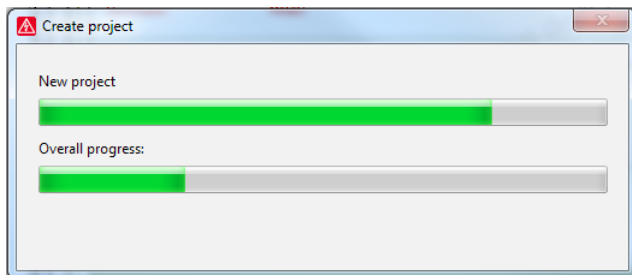


Fig. 3.12 The new project is being generated

Once EPLAN has successfully generated the data for the new project, the **Project properties** dialog is displayed. You do not necessarily need to edit these at the moment. This can be done later during project editing.

EPLAN opens the **Project properties** dialog with the *Properties* tab. The project properties can now be adjusted or completely changed on the *Properties*, *Structure* etc. tabs.

There is one limitation: The *structure of the pages*, located in the *Structure* tab, can no longer be changed (grayed out). It is fixed because the page structure was defined in the basic project.

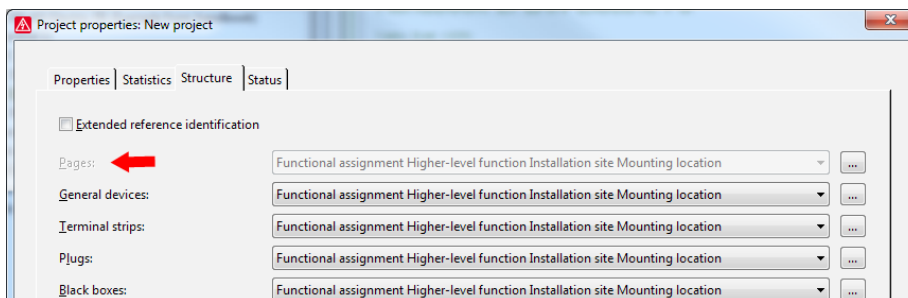


Fig. 3.13 Project properties – Structure tab

When you click **OK**, the project properties are saved and the project is immediately opened in the page navigator (it can also be opened using the **F12** key or you can display the overview with the **PAGE/NAVIGATOR** menu item). The project can now be edited.

4

The graphical editor (GED)

In EPLAN Electric P8 you mainly work in the graphical editor, also known as the GED. The graphical editor contains all the functions you need for graphical project editing. It is, as it were, the main hub of EPLAN Electric P8.

■ 4.1 Page navigator

In addition to the actual data such as parts data function templates, a project also contains individual page types, e.g. schematics, panel layout drawings or reports, such as terminal diagrams or connection lists, that can be divided into particular identifiers (structures) such as higher-level functions or mounting locations and installation sites.

To provide a certain level of clarity and make it easy to edit the page properties, EPLAN has a **page navigator**. You open the **page navigator** via the **PAGE/NAVIGATOR** menu or via the F12 key. As is usual in EPLAN, the **page navigator** offers a choice between tree structure or list view.

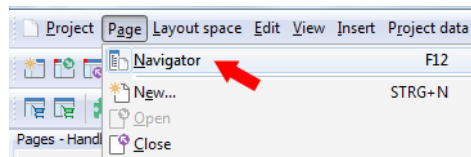


Fig. 4.1
Opening the page
navigator

Page navigator Tree/list

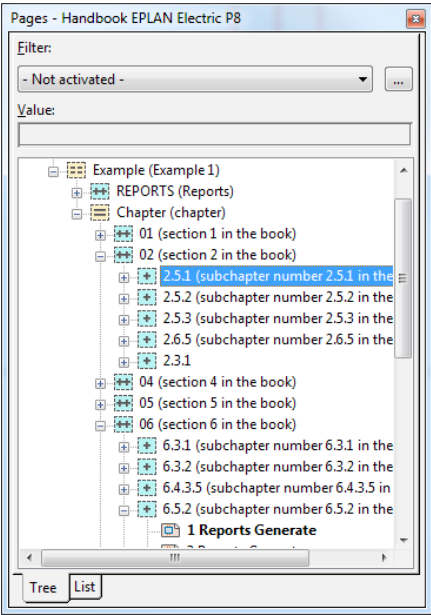


Fig. 4.2 Tree structure representation

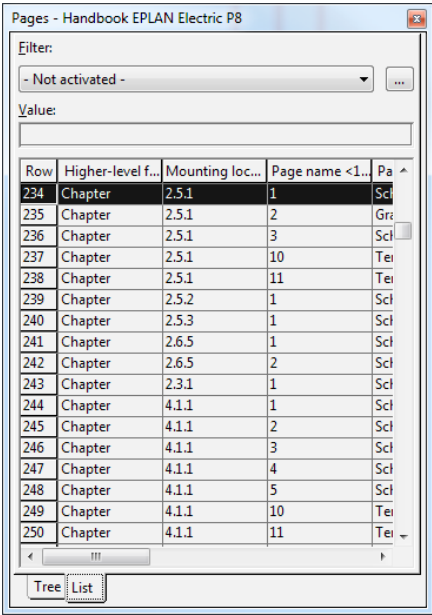


Fig. 4.3 List representation

The tree view is ideal for providing a visual overview of all pages in a project. The small symbols preceding the names graphically differentiate identifiers and pages. You can define the display sequence via the parameter in the **OPTIONS/SETTINGS/PROJECTS [PROJECT NAME]/DISPLAY/PROJECT STRUCTURE (PAGES)** menu.

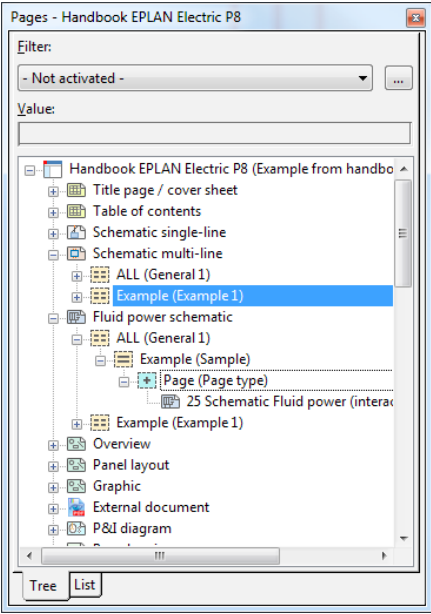


Fig. 4.4 Structured by page type

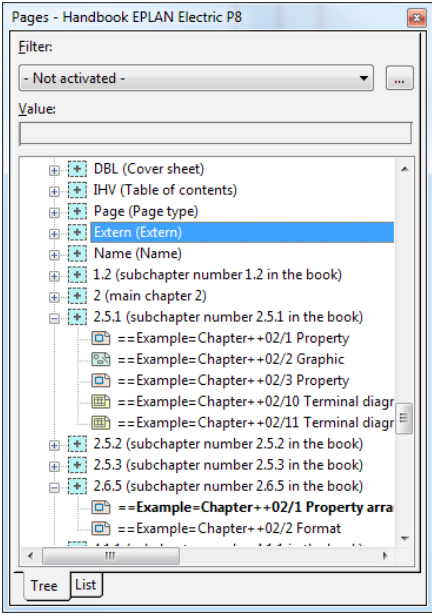


Fig. 4.5 Structured by mounting location

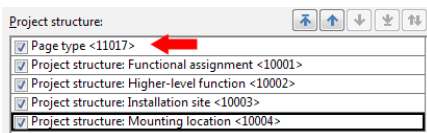


Fig. 4.6 Project setting by page type

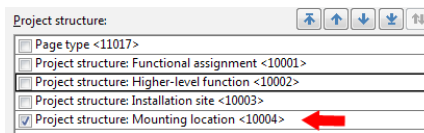


Fig. 4.7 Project setting by mounting location

This only affects the visual display of the pages in the **page navigator**. The sorting of the identifiers (higher-level function, mounting location, installation site, etc.) is defined as usual in *structure identifier management*, i.e. whether (e.g.) higher-level function = A11 is sorted before or after higher-level function = A32.

4.1.1 Page types

EPLAN offers a range of different page types for specific purposes. In EPLAN, small symbols make these page types easier to be recognized at a glance.

EPLAN distinguishes between logical and graphical pages.

Examples of logical pages are, for example, *Schematic multi-line*, *Schematic single-line* and *P&I diagram*. EPLAN examines these page types for logical information and evaluates them accordingly (cross-references, etc.).

The *Graphical* page type or the *model view*, on the other hand, are purely graphical (non-logical) pages that initially do not contain logical information.

EPLAN also makes a distinction between pages that can be edited (interactive pages) and pages that are generated (automatic pages). The *Panel layout* page type is an example of an interactive page and the *Terminal diagram* page type is an example of a generated (automatic) page.

In the page navigator, all page types in EPLAN are identified by small symbols, as described above.

The exception is that there is no page type in the *layout space*. The layout space is the basis of the 3D representation of enclosures or other components used for the panel layout; here it only provides a view of the 3D data and its further processing.

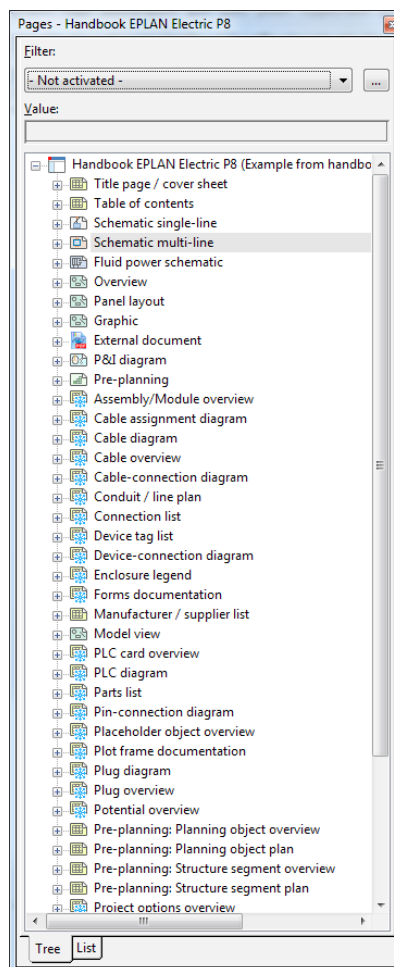


Fig. 4.8
Page types

Icon	Page type	Possible use	Interactive/ automatic
	Schematic multi-line	Multi-line schematics	Editing possible
	Schematic single-line	Single-line representation	Editing possible
	Fluid power schematic	Fluid power schematics	Editing possible
	P&I diagram	P&I diagrams	Editing possible
	Graphic	Graphical representation without logic	Editing possible
	Overview	PLC	Editing possible
	Topology	Routing path networks (2D)	Editing possible
	Pre-planning	Graphical pre-planning (no connection points and no connections possible)	Editing possible
	Table of contents	List of pages	Automatically generated page
	Manufacturer/supplier list	List of manufacturers/suppliers	Automatically generated page
	Title page/cover sheet	Cover sheet	Automatically generated page
	Terminal diagram	List of terminals	Automatically generated page
	Terminal-connection diagram	List of terminals with more graphical options	Automatically generated page
	Terminal line-up diagram	Shows terminals in a structure list, for example	Automatically generated page
	Terminal-strip overview	Overview of all terminal strips	Automatically generated page
	Cable diagram	List of a cable	Automatically generated page
	Cable assignment diagram	Representation of structure for cables with further information	Automatically generated page
	Cable-connection diagram	List of cables with more graphical options	Automatically generated page
	Cable overview	Overview of all cables	Automatically generated page
	Plug diagram	List of plug/female pin contacts	Automatically generated page
	Pin-connection diagram	List of plugs/female pins with more graphical options	Automatically generated page
	Plug overview	Overview of all plugs/female pins	Automatically generated page
	Panel layout	For the structure of mounting surfaces, such as enclosures, control panels	Editing possible

Icon	Page type	Possible use	Interactive/ automatic
	Enclosure legend	List of equipped devices of mounting surfaces	Automatically generated page
	Model view	2D drawings of 3D models generated on the basis of a layout space	Editing possible
	Connection list	List of connections	Automatically generated page
	Device tag list	List of devices	Automatically generated page
	Parts list	List of individual parts	Automatically generated page
	Summarized parts list	List of summarized parts, etc.	Automatically generated page
	PLC diagram	List of PLC connection points	Automatically generated page
	PLC card overview	PLC overview representation	Automatically generated page
	Device connection diagram	List of device connection points	Automatically generated page
	Symbol overview	Overview of symbols	Automatically generated page
	Potential overview	List of potential information	Automatically generated page
	Structure identifier overview	Overview of structure identifiers	Automatically generated page
	Revision overview	List of revision statuses	Automatically generated page
	Forms documentation	Overview of forms	Automatically generated page
	Plot frame documentation	Overview of plot frames	Automatically generated page
	Project options overview	List of project options	Automatically generated page
	Placeholder object overview	List of placeholder objects	Automatically generated page
	Topology: Routing path list	Output of routing paths and their data	Automatically generated page
	Topology: Routing path diagram	Output of routing path with cables, lines	Automatically generated page
	Topology: Routed cables/connections	Overview of routed cables, connections	Automatically generated page
	Pre-planning: Structure segment overview	List of structure segments	Automatically generated page
	Pre-planning: Structure segment diagram	List of structure segment properties	Automatically generated page

Icon	Page type	Possible use	Interactive/ automatic
	Pre-planning: Planning object overview	List of planning objects	Automatically generated page
	Pre-planning: Planning object diagram	List of planning object properties	Automatically generated page
	Pre-planning: Segment template overview	List of segment templates	Automatically generated page
	Pre-planning: Segment template plan	List of segment template properties	Automatically generated page
	Assembly/Module overview	Overview of assemblies and modules used	Automatically generated page
	Distributed device list	List of devices and associated, cross-referenced functions	Automatically generated page
	Bundle/conduit plan	List of bundles/conduits	Automatically generated page
	External document	Integration of external documents	Editing possible
	Logocad TRIGA drawing	Not used anymore; may still exist for legacy reasons	Editing possible

4.1.2 The popup menu in the page navigator

Page editing (but not graphical page editing), is done via the **PAGE** menu or via the popup menu in the page navigator.

Page menu and page navigator

You access the *popup menu* functions via the right mouse button in the **PAGE/NAVIGATOR** menu.

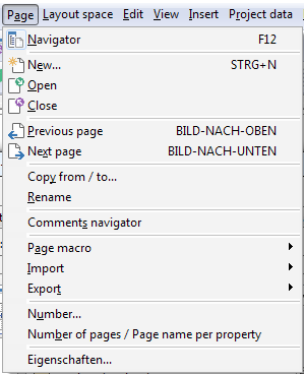


Fig. 4.9 Page menu

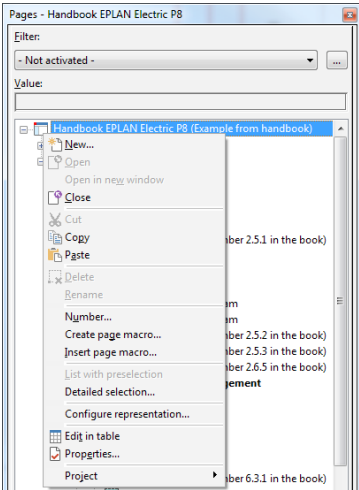


Fig. 4.10 Popup menu

4.1.2.1 Creating new pages

Usually the pages of a project are created from scratch (leaving aside copying for the time being). This can be done using the **CTRL + N** shortcut key or using the **PAGE/NEW** menu. The procedure is the same when using the popup menu, except that in this case you select the **NEW** menu entry. The **NEW PAGE** dialog then opens.

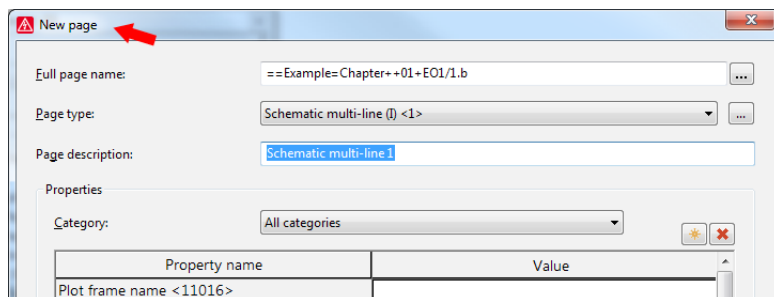


Fig. 4.11 New page dialog

There are now two ways to integrate the new page into an existing page structure. You can define the full page name in the *Full page name* field. You can either manually change the existing entry or enter a completely new value with a prefix for the structure identifier. Alternatively, you can use the  button to call up the **FULL PAGE NAME** dialog.

EPLAN opens the **FULL PAGE NAME** dialog, whereby every identifier (if used in the project's structures) has its own input field and own  selection button, which can be used to branch to the selection dialog for the selected identifier.

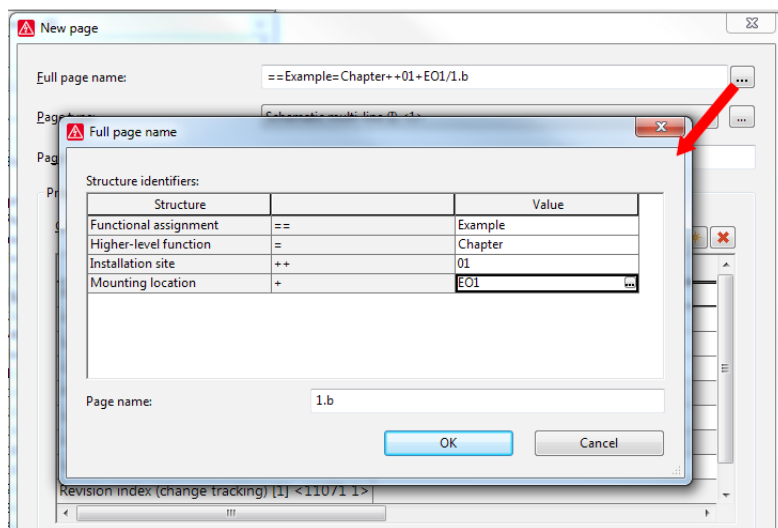


Fig. 4.12 Full page name dialog

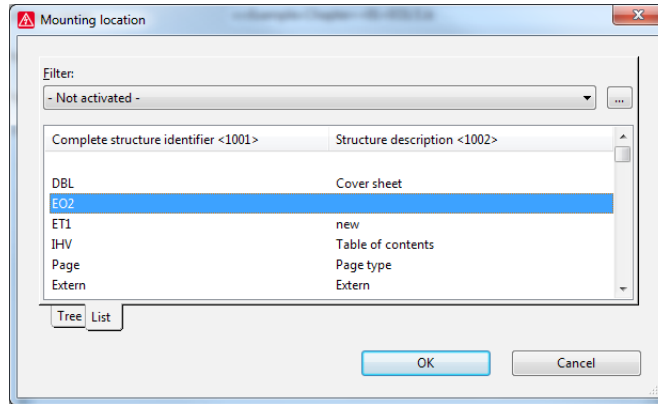


Fig. 4.13 Identifier dialog

Depending on the selected page structure, you can now click to select a *higher-level function, mounting location, etc.* Clicking OK applies the selected identifiers. This dialog, too, allows for switching between the tree and list views.



NOTE: A new identifier can also be directly entered into the [Identifier type – Value] input field in the FULL PAGE NAME DIALOG.

But keep in mind: EPLAN saves new identifiers in the order defined in the settings. This setting is under OPTIONS/SETTINGS/USER/DISPLAY/IDENTIFIER. There are no other query dialogs.

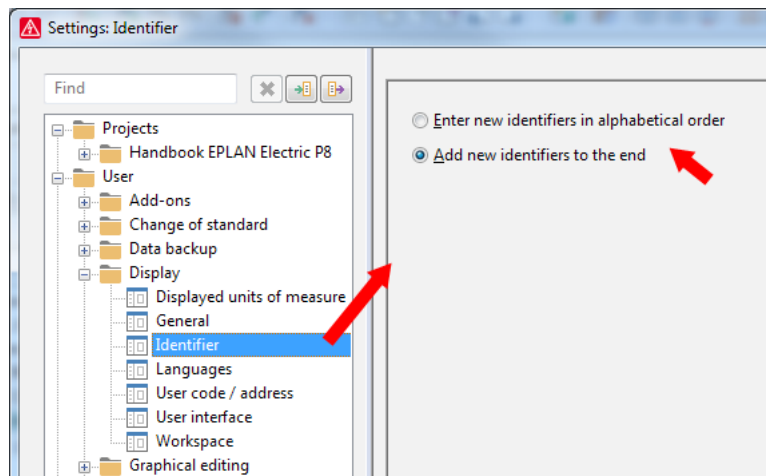


Fig. 4.14 Connection settings

Sorting can be performed manually or semi-automatically later in structure identifier management.

If all settings and entries have been entered in the **NEW PAGE** dialog and any missing structure identifiers have been created, the dialog can be closed. EPLAN will create the new page and then show it in the page navigator in the chosen structure.

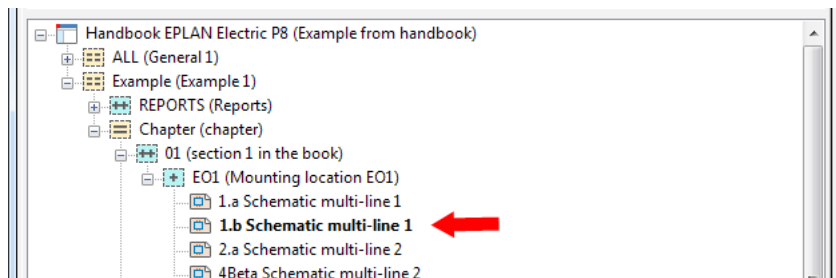


Fig. 4.15 Newly created page

4.1.2.2 Open pages

As the name of the function indicates, the **PAGE/OPEN** menu item opens a page. This is nothing new. The **OPEN IN NEW WINDOW** menu item in the page navigator popup menu is more interesting and useful. This menu item allows you to open several pages or even open the same page **several times**.

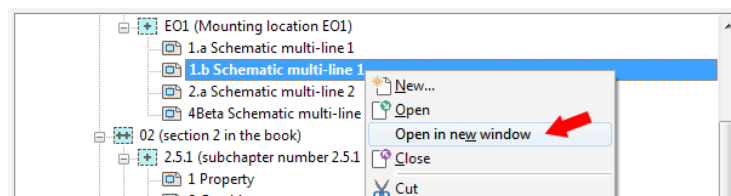
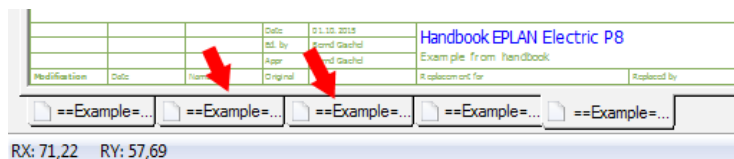


Fig. 4.16 Opening in new window

The procedure is simple. In the page navigator, you select the page or pages (it is possible to open several pages at once), right click to open the popup menu and select **OPEN IN NEW WINDOW**. You can also use the **CTRL + ENTER** keyboard shortcut. EPLAN then opens all selected pages.



Function: Open in new window

Fig. 4.17 Several open pages

If the **WORKBOOK** menu item in the **VIEW** menu is activated, then all opened pages are displayed in Excel style (tabbed page list) at the bottom of the workspace. You can then use the mouse to switch back and forth between these pages – or the **CTRL + TAB** keyboard shortcut to sequence through all the pages. With three pages, the sequence starts at page 1 followed by page 2, page 3, back to page 1, etc.

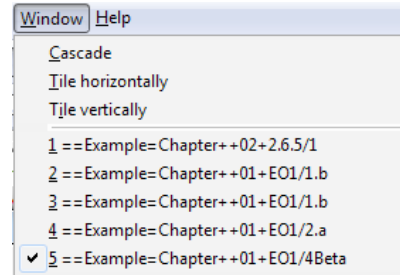


Fig. 4.18
Window menu

If several pages are open, you can use the **WIN-DOW** menu to go directly to the desired page.

To close open pages, simply select the pages in the page navigator and in the **PAGE** menu, select **CLOSE**. EPLAN then closes all selected pages.

4.1.2.3 Copy pages

When you create new pages, they are created without any page content (devices, graphics, etc.). To create a new page that uses the content of an existing page, you need to copy one or more pages. There are several ways to do this in EPLAN. You can use the standard Windows functions like **CTRL + C** (copy page) and **CTRL + V** (paste page), or the **PAGE** menu (or the page navigator popup menu).

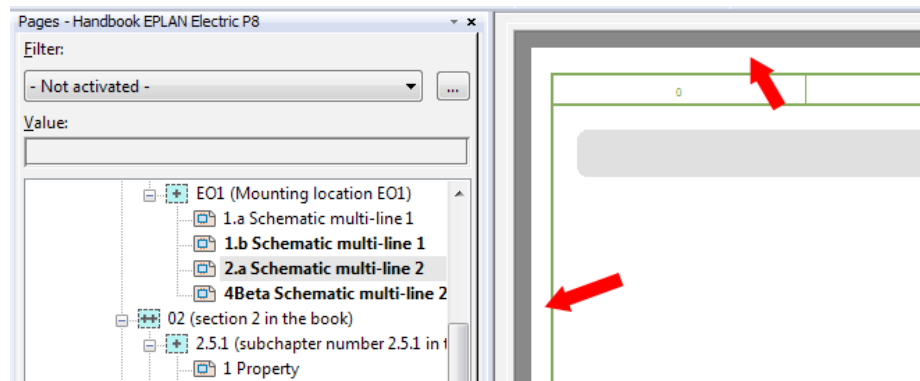


Fig. 4.19 Using **CTRL + M** to select a page

Use **CTRL + M** to select a page (outside the page navigator) in the graphical editor in order to copy it. EPLAN then draws a thick gray border around the page (to show it is selected). When you press **CTRL + C** (standard Windows function for copying), EPLAN copies the page to the clipboard. Pressing **CTRL + V** (paste) retrieves the copied page from the clipboard and pastes it to a different location. EPLAN then opens the **ADAPT STRUCTURE** dialog.